

Virus Evolution and Emergence



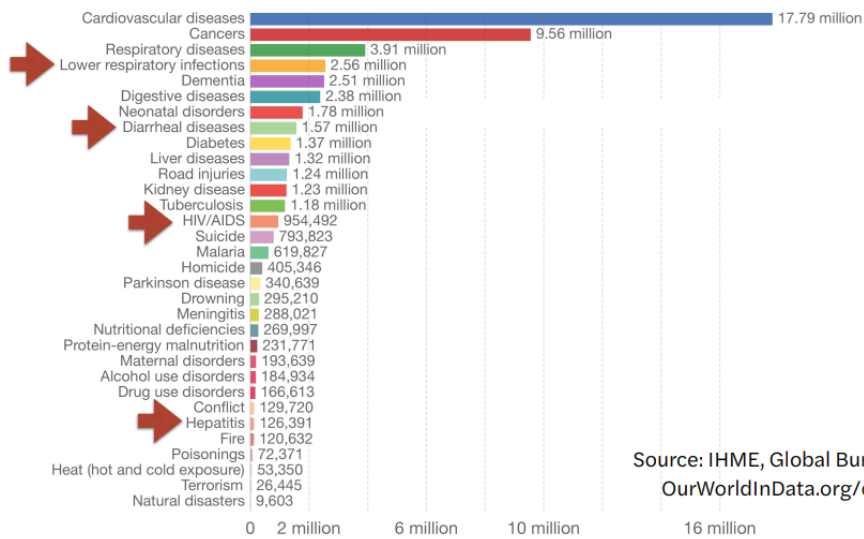
By: Ahmed El-Shamy, DVM, Ph.D

On: April 14, 2026

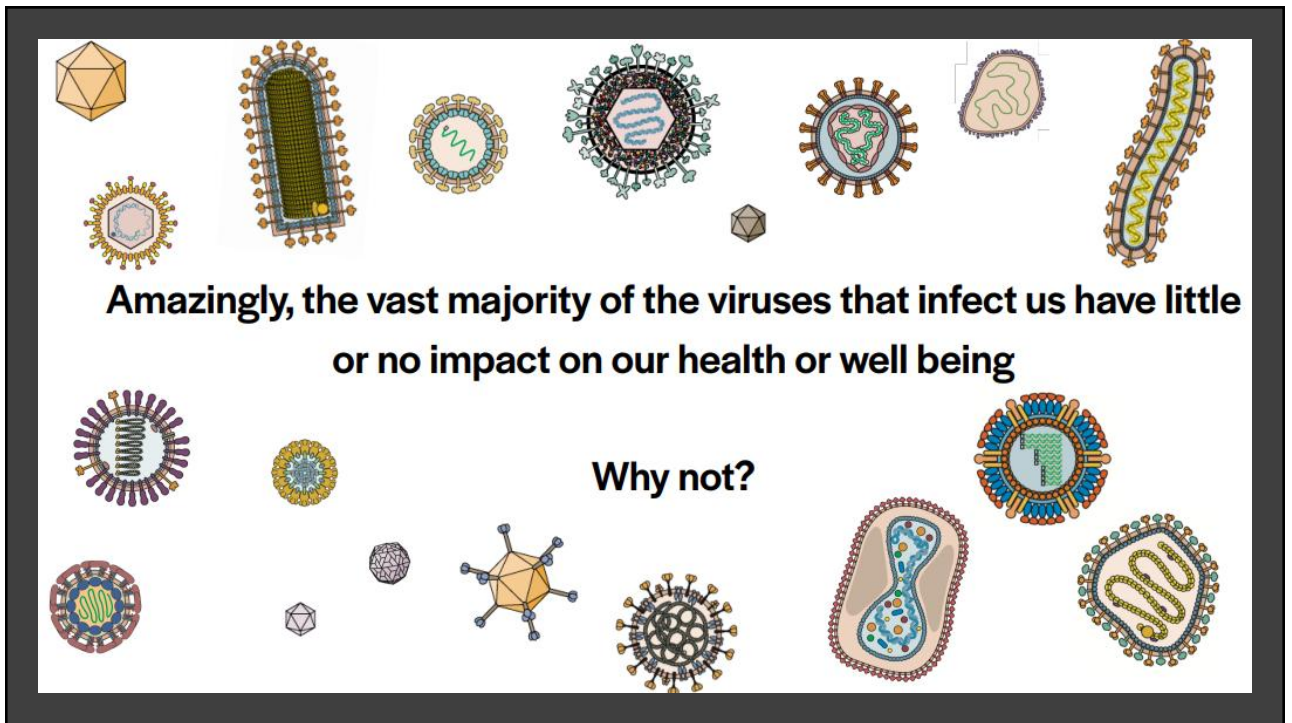
Lecture Objectives

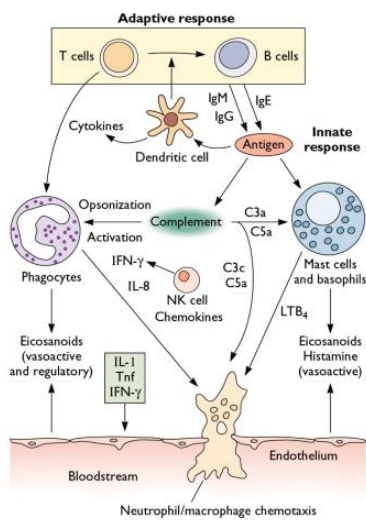
- 1- Understand the strategies of virus evolution.
- 2- Understand the nature of virus emergence.
- 3- Discuss examples of emerging viral diseases.
- 4- Identify how AI could help us prepare for the next pandemic.

Causes of 2017 global deaths



Source: IHME, Global Burden of Disease
OurWorldInData.org/causes-of-death



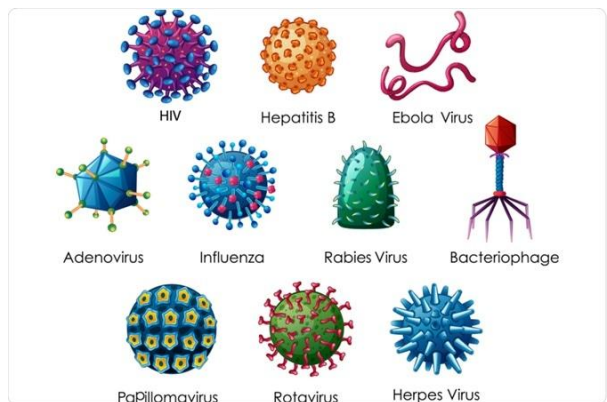


We have an amazing immune system

Our immune system can manage most viral infections

What is a virus?

- The word virus comes from the Latin for a poisonous liquid. It was originally used to describe any infectious agent
- The genetic material of a virus enters a host cell (called infection) and directs the production of the building blocks of new virus particles (called virions).
- The new virions produced in a host cell then transport the viral genetic material to another host cell or organism to carry out another round of infection.

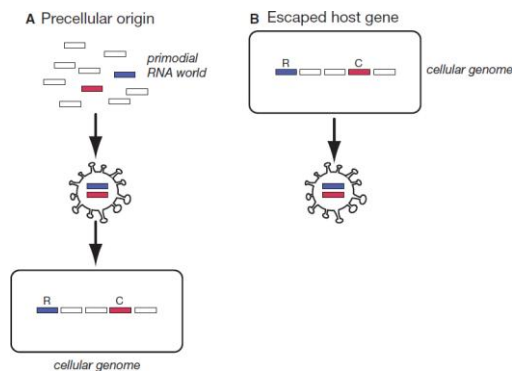


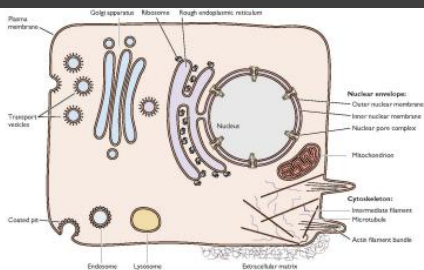
How and when viruses first evolved!

- Determining exactly how and when viruses first evolved is perhaps the most challenging. The main hindrance to progress in this area is that viruses likely originated so long ago, perhaps even before the first cellular species.
- Two theories currently dominate the understanding of viral origins:

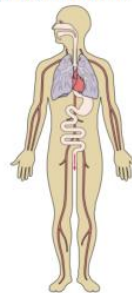
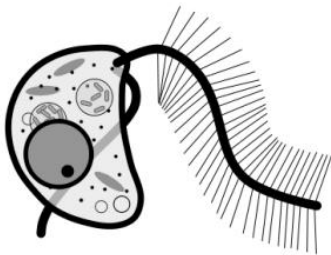
A. Viruses have a precellular origin, such that they are billions of years old, and may have even contributed to some of the fundamental architecture of the first cells.

B. Viruses evolved after the first cellular organisms as “escaped genes” that acquired capsid proteins and the ability to replicate autonomously





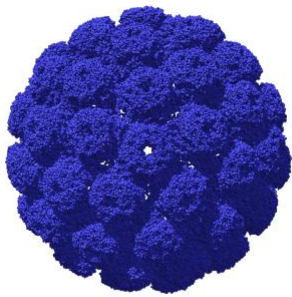
As viruses are obligate molecular parasites, every solution must reveal something about the host as well as the virus



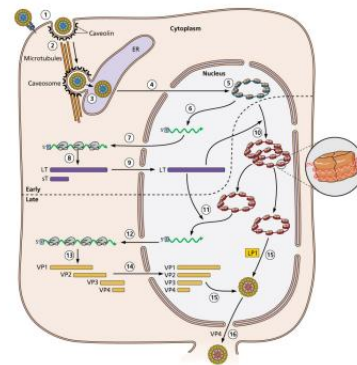


Are viruses alive?

A virus is an organism with two phases



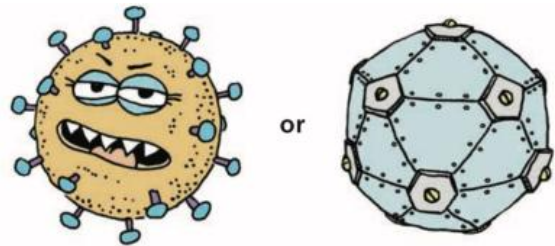
virion (infectious particle)



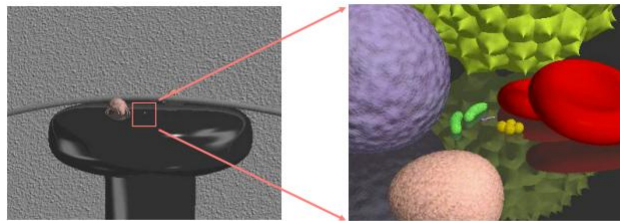
infected cell

Be careful: Avoid anthropomorphic analyses

- Viruses do NOT think!
- They do not achieve their goals in a human-centered manner.
- Viruses are passive agents!



**How many viruses can fit on the
head of a pin?**



2 mm = 2000 microns

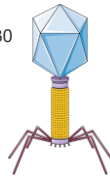
500 million rhinoviruses

When you sneeze, you fire an aerosol that contains enough viruses to infect thousands.

**How many viruses are on the
earth?**

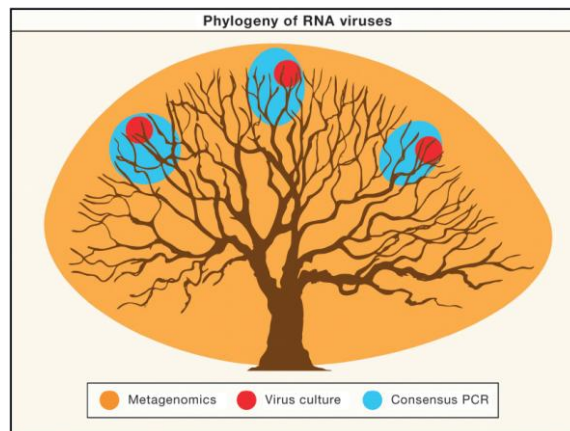
How many viruses are on the earth?

- We live and prosper in a cloud of viruses.
- We regularly eat and breathe billions of virus particles.
- We carry viral genomes as part of our own genetic material.
- More than 10^{30} bacteriophage particles in the world's waters!
- The length of a head to tail line of 10^{30} phages is 100 million light years!



Welcome to the Virosphere:
(the world of virus diversity)

“The number of undetected viruses is enormous such that we are still only scratching the surface of the virosphere!”



Different analytical techniques to explore the virosphere. A schematic tree is used to represent the total phylogenetic diversity of RNA viruses (although the same scheme also applies to DNA viruses). The *colored circles* depict the extent of viral diversity that can be discovered using cell culture (*red*), consensus PCR (*blue*), and metagenomics, such as RNA sequencing (meta-transcriptomics, *orange*).

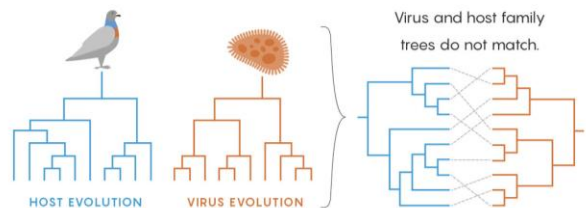
Virus Evolution

Modern virology has provided a window on the mechanisms of evolution

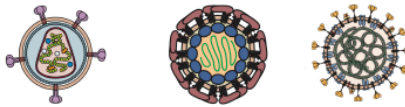
- As host populations grow and adapt, virus populations are selected that can infect them.
- *New viral populations emerge every day.*
- It also works the other way.
- *Viral populations can be significant selective forces in the evolution of host populations.*
- If a host population cannot adapt to a lethal virus infection, the population may be exterminated.

Cross-species Transmission

If viruses evolve mostly by jumping into new host species unrelated to their old ones, then the evolutionary relationships among hosts and among viruses will be mismatched. Research finds that this pattern is most common.



The public is constantly confronted with the reality of viral evolution

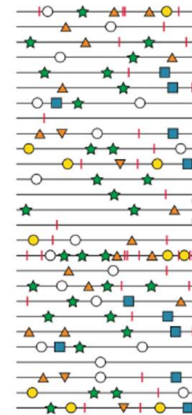


- New viral diseases: AIDS, West Nile virus disease in the US, hepatitis C, Ebolavirus disease, Zika virus disease, COVID-19
- Regular bouts every year with influenza and common cold viruses
- Drug resistant HIV-1

**Simple fact: virus evolution is faster
than many can comprehend**

Four main drivers of virus evolution

- Large numbers of progeny
- Large numbers of mutants
- Quasi-species effects
- Selection



Virus-infected cells produce large numbers of progeny

Virus in plasma	HBV	HIV
Half-life	24 h	6 h
Daily turnover	50%	90%
Total production in blood	$>10^{11}$	$>10^9$

The interface of host defense and virus replication is fertile ground for selection and evolution

Replicating viruses produce large numbers of mutant genomes

- Evolution is possible only when mutations occur in a population
- Mutations are produced during copying of any nucleic acid molecule

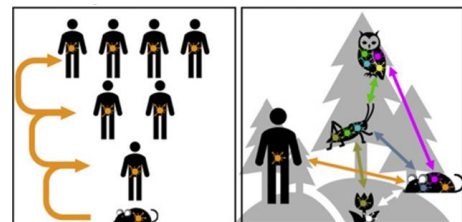
Viral genomes are always mutating!

The role of mutation in virus evolution

- One of the strongest arguments that viruses are alive is that they are subject to the same forces that shape the evolution seen in cellular species including mutation, natural selection, genetic drift and recombination.
- The simplest way to examine the role of mutation in virus evolution is to measure the rate of its occurrence.
- For some viruses, especially RNA viruses, mutation occurs at a pace that greatly exceeds that observed in other organisms and determines much of their evolutionary “behavior.”

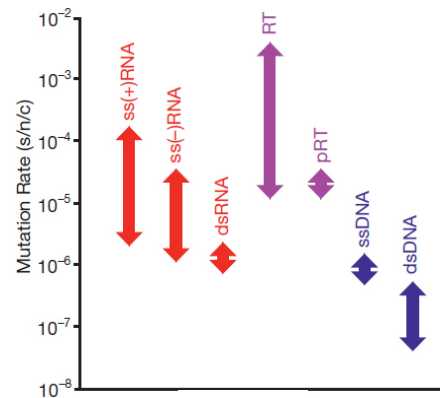


Emergence



The role of mutation in virus evolution

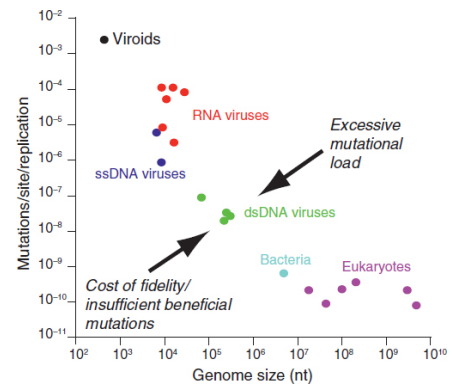
- Mutation rate estimates the intrinsic error dynamics of the RNA or DNA polymerases used in viral replication.
- For RNA viruses that replicate with a RdRp, an enzyme that lacks a proofreading or repair function, this equates to mutation rates that are usually around one per genome, per replication.
- The lower mutation rates observed in DNA viruses clearly reflect the higher fidelity of the DNA polymerases employed in their replication cycle.
- In general, most estimates of substitution rates suggest a fundamental division in virus evolution in which RNA viruses evolve rapidly and DNA viruses evolve slowly.



The large-scale variation in mutation rates among viruses. The range of variation in mutation rates is shown for the seven classes of viruses classified under the Baltimore scheme. ss, single-strand; ds, double-strand; +/- genome polarity; RT, retroviruses; pRT, para-retroviruses.

The role of mutation in virus evolution

- There is a strongly inverse relationship between mutation rate and genome size that applies to all living systems.
- Although they utilize related DNA polymerases, mutation rates in large DNA viruses are higher than those of bacteria and eukaryotes, which may reflect an absence of the full set of repair enzymes and pathways.



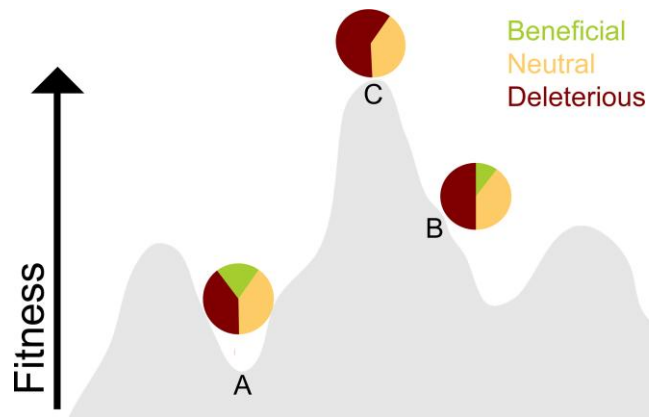
Relationship between mutation rate and genome size in diverse organisms including RNA and DNA viruses and viroids. The competing evolutionary forces that might be responsible for the limited range of observed mutation rates and genome sizes are shown.

The role of mutation in virus evolution

- Viral mutation rates are the result of an evolutionary trade-off between replication rate and replication fidelity.
- Rapid replication is selectively advantageous for a virus but comes at the cost of lower replication fidelity.
- However, this evolutionary trade-off will vary depending on virus type and the duration of infection, for example, between viruses associated with either acute or chronic infections.



The role of mutation in virus evolution



A fitness landscape showing three genotypes on different places on the landscape (A, B, and C) and a schematic pie chart of the distribution of mutations available to each genotype. The genotype at A is not well adapted to the environment (far from a fitness peak) so has a larger fraction of mutations that would be beneficial. The genotype at B is more fit than A and is closer to a fitness peak, so it has a smaller fraction of beneficial mutations than that at A. The genotype at the fitness peak C does not have any way to become more fit on this landscape and thus has no beneficial mutations available to it.

Error threshold

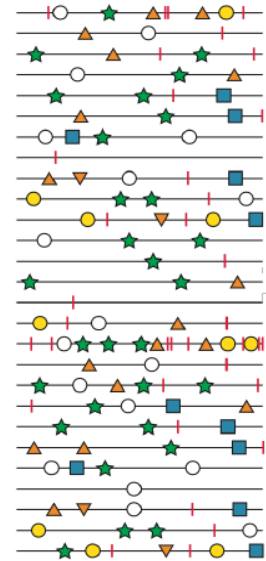
- Mutation is a powerful advantage, but selection and survival balances mutation rate.
- This limit is called the error threshold.
 - *Exceed it*: loss of infectivity
 - *Below it*: cannot produce enough mutations to survive selection.
- RNA viruses: evolve close to their error threshold.
- DNA viruses: evolve far below their error threshold.
- What is the number of all possible mutations of a viral genome?
- Sequence comparisons of several RNA virus genomes have demonstrated that well over half of all nucleotides can accommodate mutations

The fundamental properties of viruses constrain and drive evolution

- Despite many rounds of replication, mutation, selection, we can recognize a herpesvirus or influenza virus genome by sequence analysis.
- Viral populations often maintain master or consensus sequences, despite opportunities for extreme variation.
- How is stability maintained?

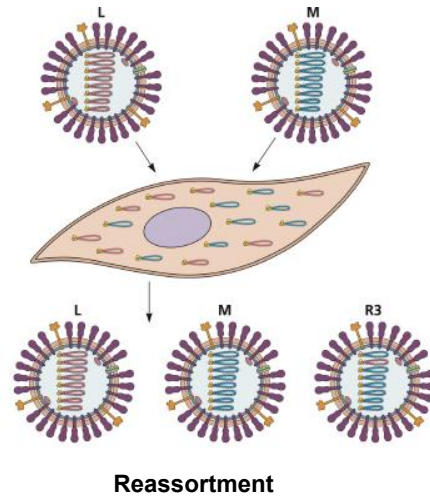
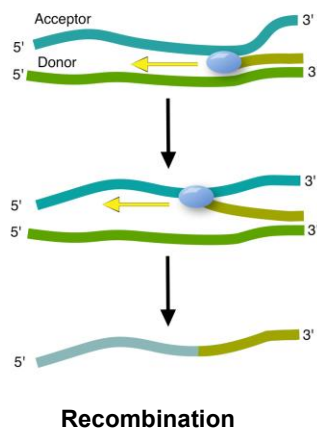
Viral quasispecies

- Virus populations exist as dynamic distributions of nonidentical but very related replicons, called quasispecies. Some prefer the term mutant swarm or cloud.
- Viral infections are initiated by a population of genomes, not a single virus genome.
- The large number of progeny produced are products of selective forces inside the host.
- The survivors that can re-infect a new host reflect the selective forces outside the host.



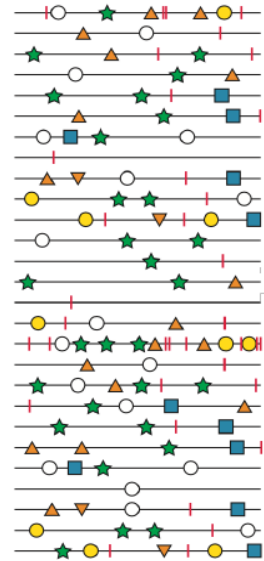
Viral quasispecies

Variation further generated by recombination and reassortment



Selection

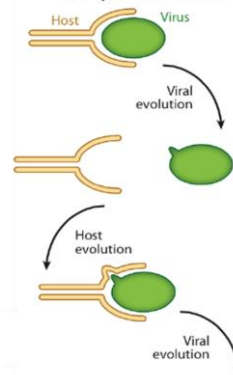
- Survival of the fittest: A rare genome with a particular mutation may survive a selection event, and this mutation will be found in all progeny genomes.
- Survival of the survivors: However, the linked, but unselected mutations, get a free ride.
- Consequently, the product of selection after replication is a new, diverse population that shares the selected and closely linked mutations



An example of selection: genetic shift & drift

- Selection of viral mutants resistant to elimination by antibodies or cytotoxic T cells inevitable when sufficient virus replication occurs in an immunocompetent individual.
- **Drift** - diversity arising from copying errors and immune selection - may occur each time a genome replicates.
- **Shift** - diversity arising after recombination and reassortment.

Host-virus arms race *Red Queen conflicts*



An evolving host coming to an equilibrium with the pathogen

Transmission of the virus to a new host is a major selective force for viral evolution

Selection: Is virulence a positive or negative trait?

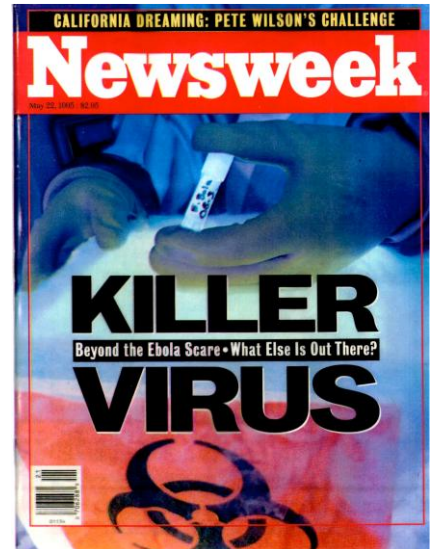
- *Positive trait*: increased virulence is a consequence of high viral loads, facilitating transmission.
- *Negative trait*: increased virulence reduces transmissibility because hosts die faster, reducing exposure to uninfected hosts.
- Probably both: there are many virulent and avirulent viruses

Emerging Viruses

"New viruses can only arise from viruses that are now in existence, not de novo"

Emerging Viruses

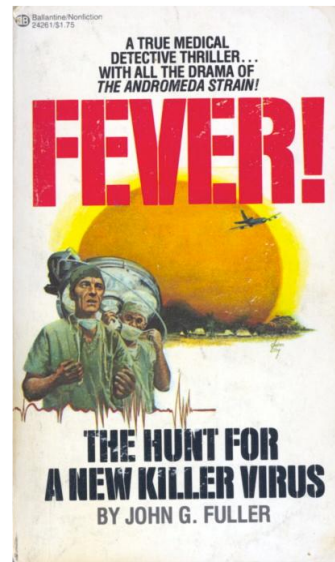
- **Emerging virus** - causative agent of a new or previously unrecognized infection.
- The term became popular in 1990s, but emerging viruses are not new.
- Since the rise of agriculture - 11,000 years ago - new infectious agents have invaded human populations because they can be sustained by numbers that were unknown before agriculture and commerce.
- new viruses can only arise from viruses that are now in existence, not de novo
- Only recently have we become good at detecting emerging viruses



Emerging Viruses

- Expanded host range with an increase in disease not previously obvious.
- Transmission of a virus from a wild or domesticated animal to humans – *zoonosis*.
- Cross-species infection may establish a new virus in the population (SIV moving from chimps to humans, SARS-CoV-2).
- Often cross-species infection cannot be sustained (e.g. Ebolavirus from bats to humans, MERS-CoV from camels to humans)

Transmission of the virus to a new host is a major selective force for viral evolution



Ecological and anthropogenic activities that promote virus emergence

- Human continuously devise new ways to encounter viruses.
- Human population growth and incredible change occurring in all ecosystems brought about by human occupation of almost every corner of the planet



Dams and water impoundments
Irrigation
Massive deforestation
Rerouting of wildlife migration patterns
Wildlife parks
Long-distance transport of livestock and birds



Blood transfusion
Xenotransplantation
Societal changes with regard to drug abuse and sex



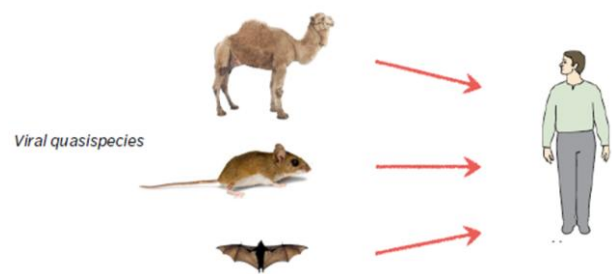
Air travel
Uncontrolled urbanization
Day care centers
Hot tubs
Air conditioning
Millions of used tires



Climate change
Natural disasters

Emerging infection needs **THREE** steps

1. Successful encounter of susceptible new host.
2. Establishment infection: replication and progeny production.
3. Transmission – spread.



Factors that led to emergence of new viruses

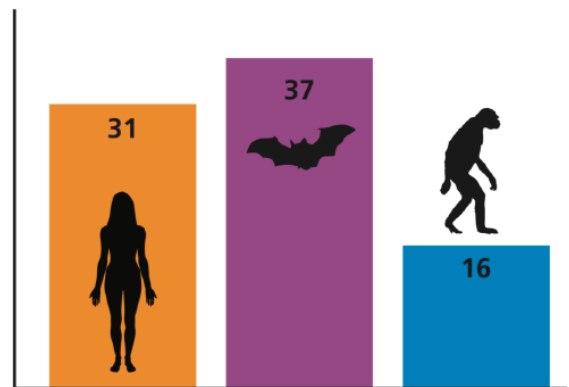
Virus	Family	Factors leading to emergence
Dengue virus 	<i>Flaviviridae</i>	Urban population density; open water storage favoring mosquito breeding (e.g., millions of used tires)
Ebola virus 	<i>Filoviridae</i>	Human contact with unknown natural host (Africa)
Hantaan virus 	<i>Bunyaviridae</i>	Agriculture: human-rodent contact during rice harvest
Hendra virus 	<i>Paramyxoviridae</i>	Proximity of fruit bats favors transmission to horses and stable workers
Human immunodeficiency virus type 1 	<i>Retroviridae</i>	Hunting and butchering of infected primates (bushmeat trade)

Influenza virus		<i>Orthomyxoviridae</i>	Reservoir in aquatic birds; expansion of bird and pig farming
Middle East respiratory syndrome (MERS) coronavirus		<i>Coronaviridae</i>	Camel husbandry, contact with humans
Nipah virus		<i>Paramyxoviridae</i>	Proximity of fruit bats, the natural reservoir, favors transmission to pigs and then to humans
Severe acute respiratory syndrome (SARS) coronavirus		<i>Coronaviridae</i>	Open-air meat markets
West Nile virus		<i>Flaviviridae</i>	Mosquito transmission from bird reservoir; global travel
Zika virus		<i>Flaviviridae</i>	Mosquitoes, global travel

Ancestral origins of human pathogens

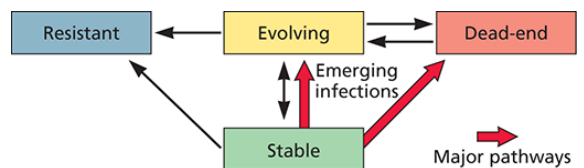
- All known types of viruses likely evolved long before humans appeared on Earth.
- All human viruses have therefore evolved from animal viruses.

Viruses belonging to # of genera

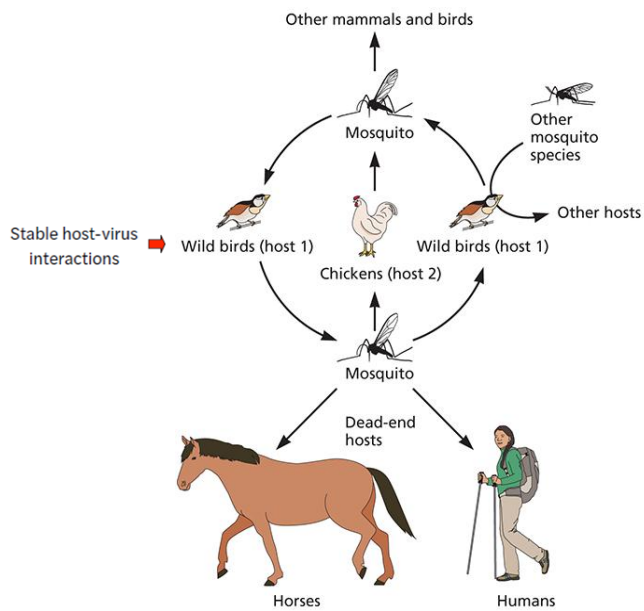


General categories of interactions between hosts and viruses

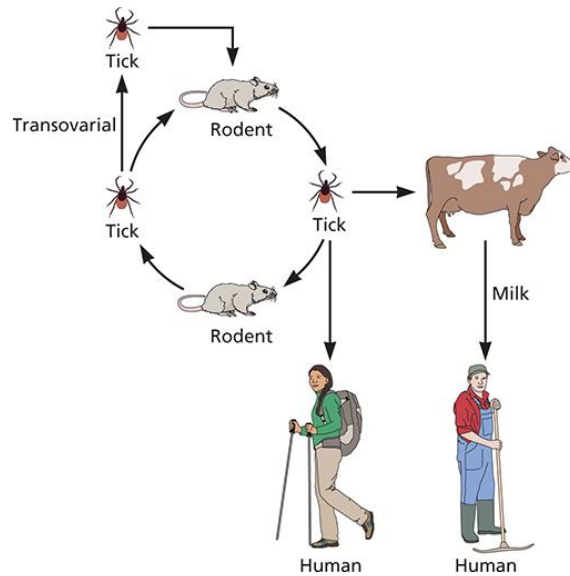
- *Stable*: maintains virus in ecosystem.
- *Evolving*: passage of virus to naive population (same or different host).
- *Dead-end*: one way passage to different species.
- *Resistant host*: infection blocked.
- *Dead-end interaction* is frequent outcome of cross-species infection. No sustained transmission from new infected host to others of the same species



Example of stable and dead-end host-virus relationships



Stable and dead-end relationships in the reproduction cycle of tick-borne encephalitis virus



Common Sources for Animal-to-Human Transmission



Rodents



Bats

Coronaviruses

Rise and fall of a zoonotic infection

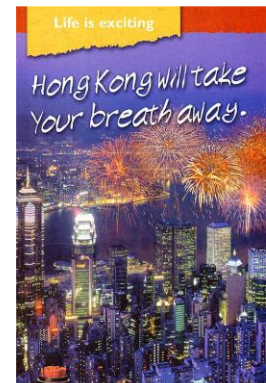
SARS (severe acute respiratory syndrome)

- Outbreak of severe atypical pneumonia, unknown etiology, Guangdong Province, China, Nov 2002.
 - 305 cases, 5 deaths
- Incubation period 2-10 days.
- Illness begins with prodrome fever, *chills*, *headache*, *malaise*, *myalgia*.
- Next phase: dry cough, shortness of breath.
- 10-20% may require mechanical ventilation



SARS

- A Chinese doctor who treated first patients traveled to Hong Kong on 21 February 2003, stayed on ninth floor of Metropole Hotel.
- He died in hospital 22 February.
- Infection spread to 10 people in hotel, who flew to Singapore, Vietnam, Canada, US before symptoms evident.
- Infection spread to 8000 people in 29 countries, 10% mortality



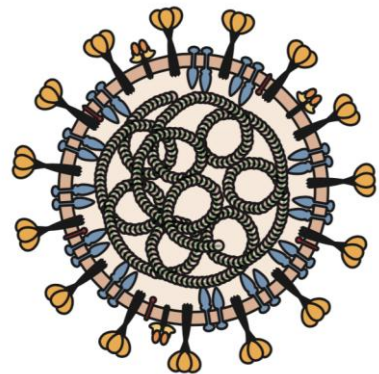
Origin of SARS-CoV

- Early Guangdong SARS cases were in handlers of animals for the exotic food market.
- Animal traders had significantly higher prevalence of anti-SARS-CoV antibodies than control groups.
- Highly diverse SARS-CoV found in horseshoe bats (*Rhinolophus*) in a cave in Yunnan Province.
- No SARS cases in Yunnan Province during outbreak.
- Direct progenitor of SARS-CoV thought to have emerged in bats, then infected civet cats which were then transported to Guangdong markets.



Middle East respiratory syndrome coronavirus (MERS-CoV)

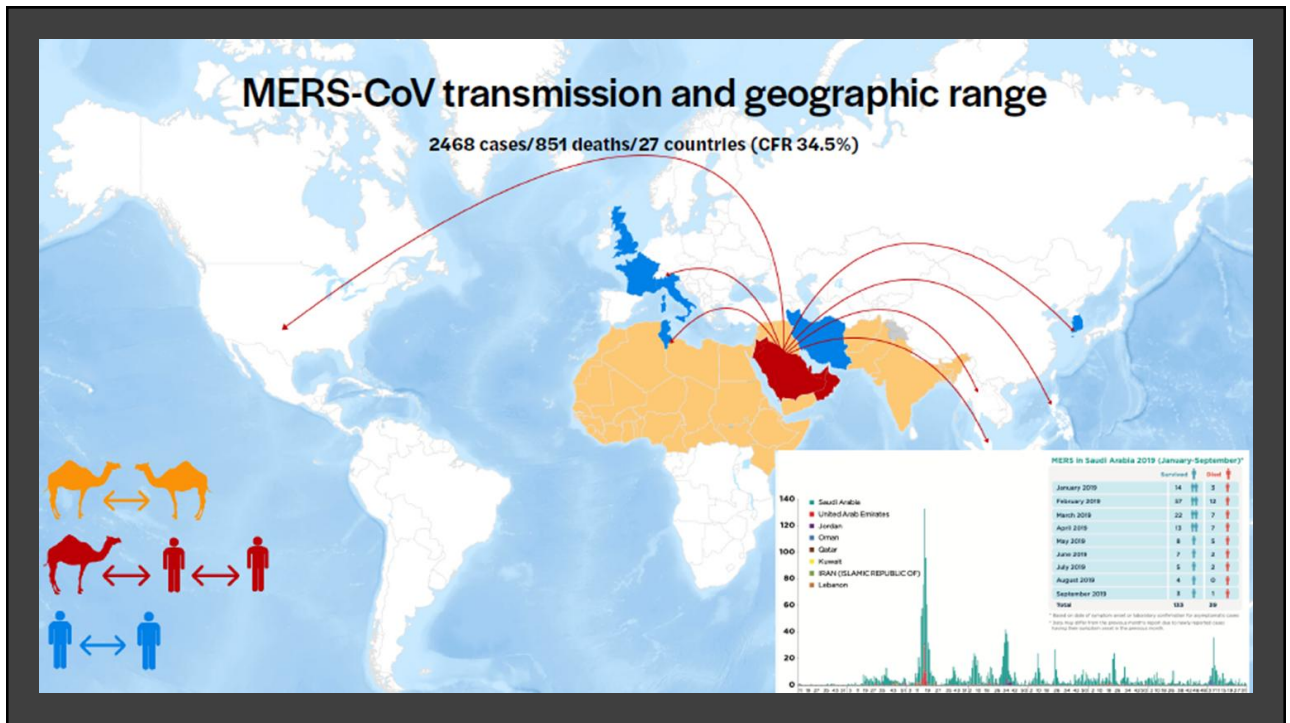
- Index case September 2012: 60 yo male patient, died of pneumonia, renal failure.
- Virus recovered, genome sequenced, binds dipeptidyl peptidase 4 receptor.
- Closely related to bat coronaviruses.



MERS-CoV

- Spread from dromedary camels (endemic in Middle East and Africa).
- Not all infections have camel source.
- How is the virus transmitted to humans?
- Virus does not transmit well among humans.
- Camel vaccine in testing.





‘Don’t go to Wuhan, don’t leave Wuhan’: Coronavirus could mutate and spread further, China officials warn

Some 440 cases confirmed so far as disease reported in Thailand, Japan, South Korea and United States

China warns virus could mutate and spread as death toll rises

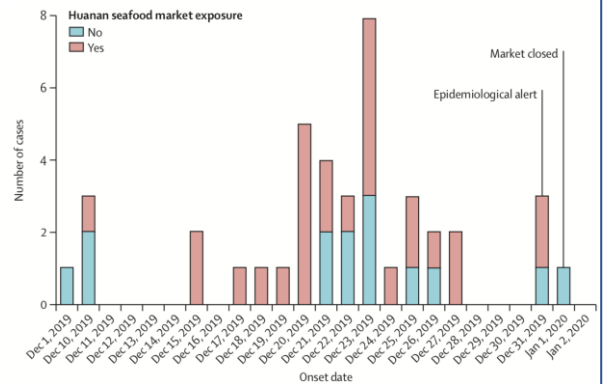
Updated / Wednesday, 22 Jan 2020 11:45



“Evidence has shown that the disease has been transmitted through the respiratory tract and there is the possibility of viral mutation” — National Health Commission Vice Minister Li Bin

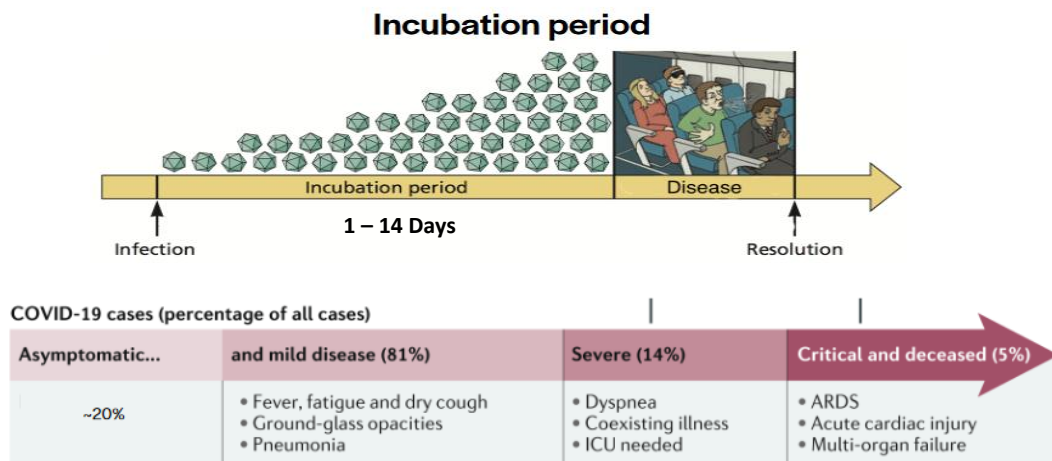
SARS-CoV-2

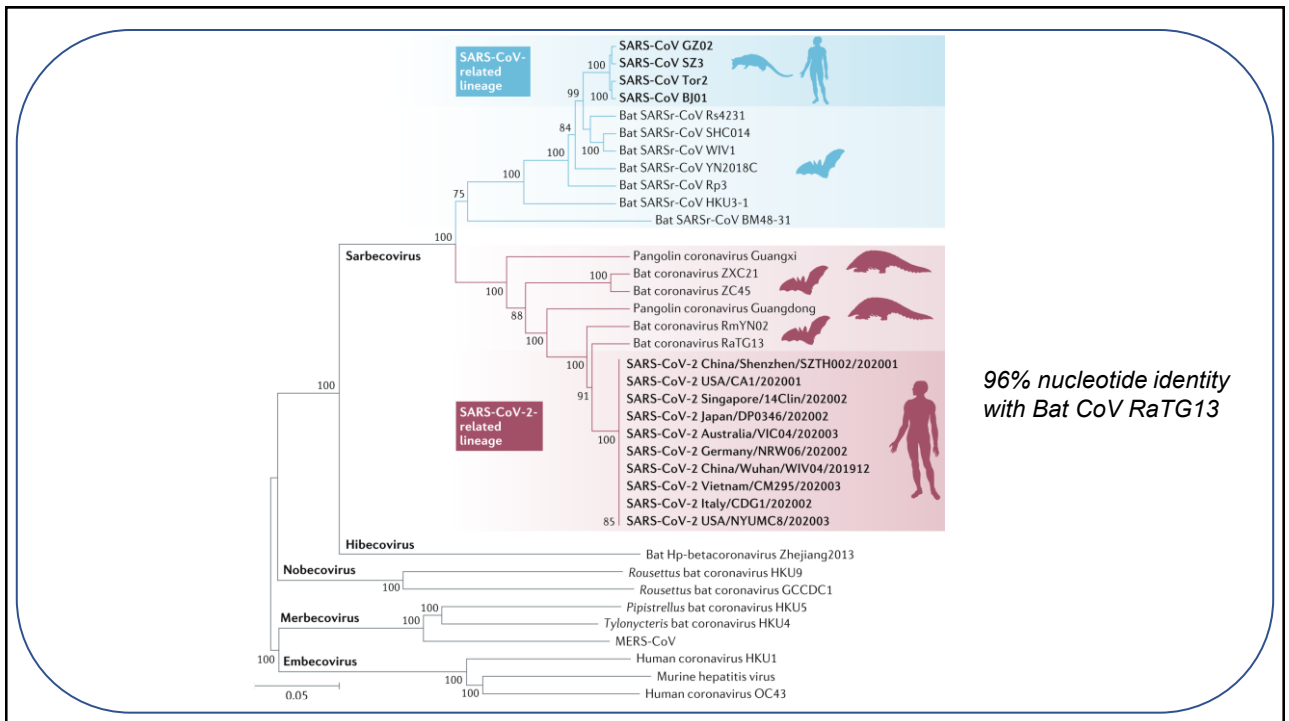
- Cluster of pneumonia cases, unknown cause, Wuhan December 2019.
- Huanan seafood market initially suspected as origin.



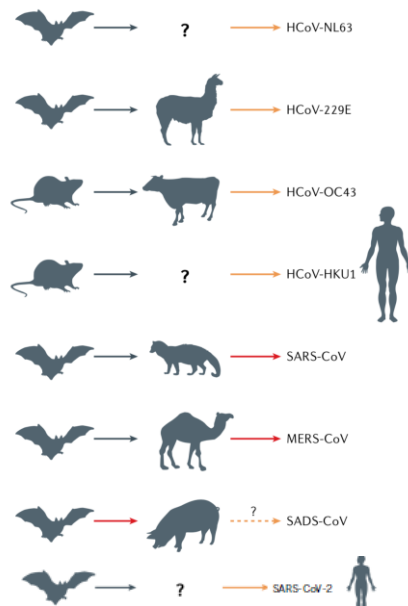
SARS-CoV-2

"As of April 2024, 700 million reported cases, 7 million deaths (0.01% CFR)"

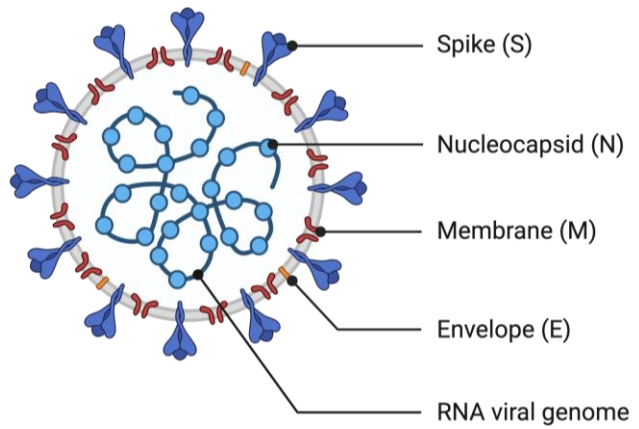
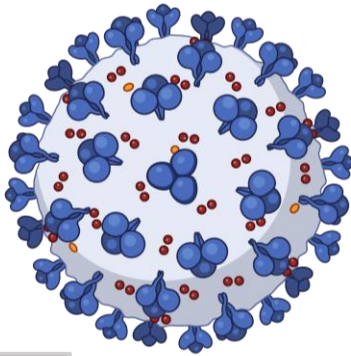




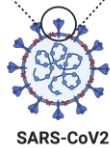
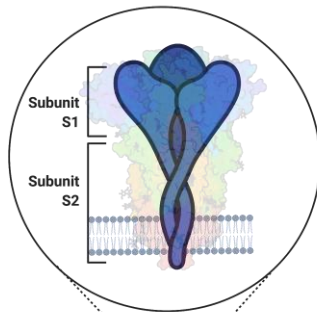
Origins of human coronaviruses - in Nature, not in a lab



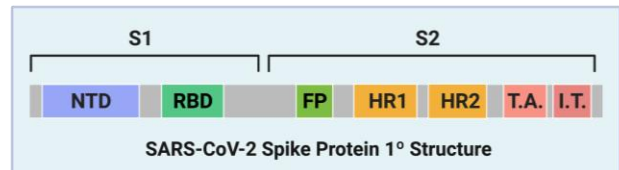
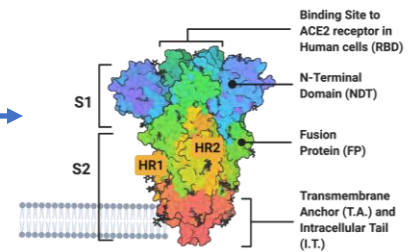
SARS-CoV-2 Structure



SARS-CoV-2 Spike Protein Structure

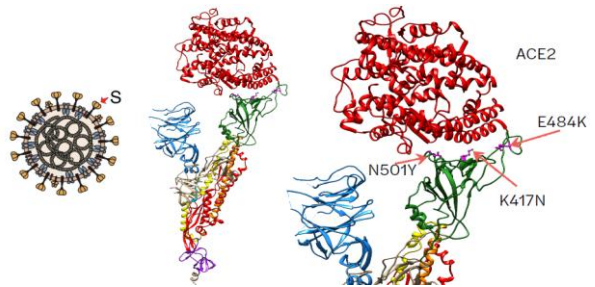


Crystallographic structure seen by electron microscopy



SARS-CoV-2 is undergoing antigenic drift

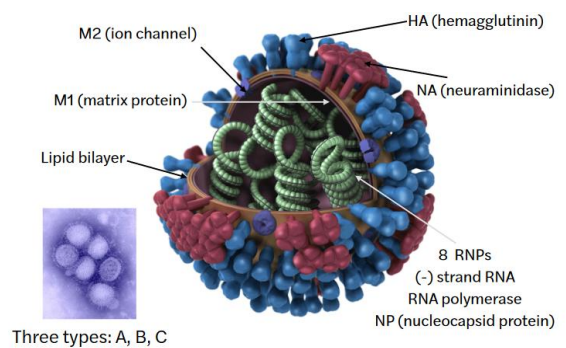
- SARS-CoV-2 variants of concern (VOC) have increased *FITNESS*.
- Fitness is the ability of the virus to reproduce in the host.
- Improved fitness can be conferred by changes in many processes, including
 - Reproduction
 - Transmission
 - Immune evasion



Influenza

Influenza Viruses

- Influenza A viruses are classified by antigenic composition, by serologic testing of HA and NA.
- Combinations of envelope proteins; HA and NA are called HxNy.
- HA is critical for viral entry to the host cell, while NA is critical for viral release.
- $x = 1-18$; $y = 1-11$.
- H1-17 can infect birds; H1, H2, H3 can infect and transmit between humans.
- the influenza A viruses that infect humans during winter 2024/2025 are labeled H1N1 and H3N2.

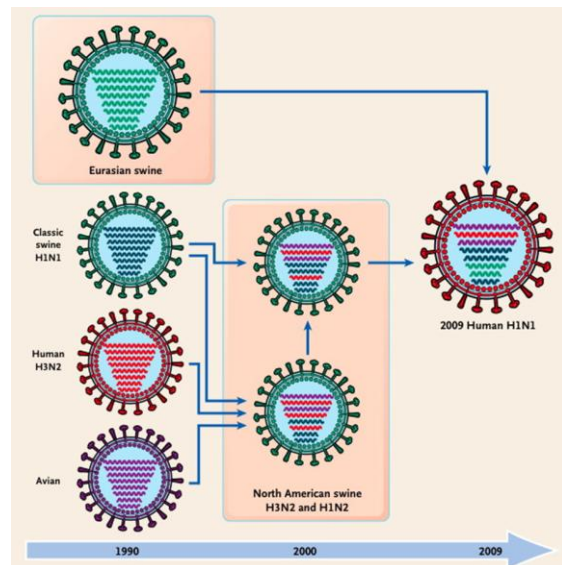
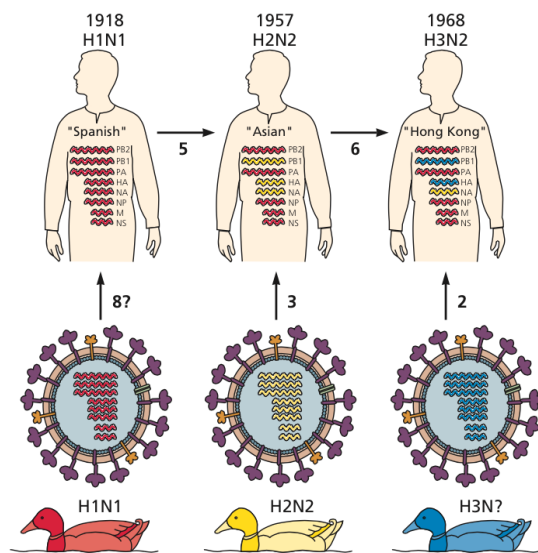


Spanish Flu 1918



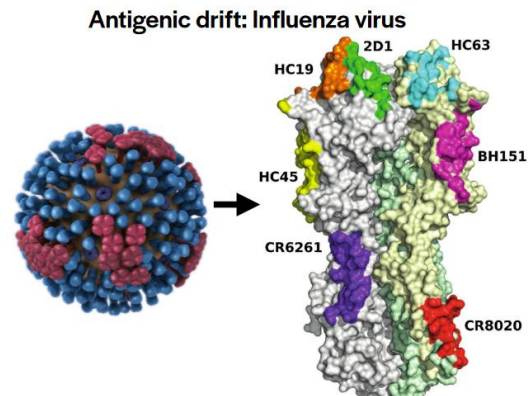
<https://www.cdc.gov/flu/pandemic-resources/1918-commemoration/historical-images.htm>

Influenza pandemics through the history



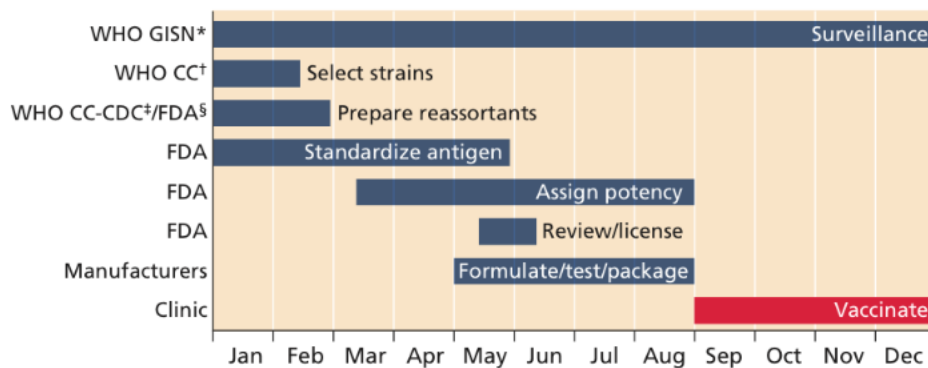
Inactivated influenza vaccine

- 3000-49000 deaths/yr in US due to influenza virus.
- The envelope proteins of influenza viruses change by antigenic drift and shift as the virus reproduces in various animal hosts around the world. Consequently, protection one year does not guarantee protection the next, which is why annual flu shots are strongly recommended.
- Vaccine: virus grown in embryonated chicken eggs, formalin-inactivated or detergent or chemically disrupted virions.
- Vaccines produced in cell culture (Flucelvax).
- A major limitation of the seasonal influenza vaccine is that it does not confer protection in all vaccinated individuals



Selecting an influenza virus vaccine

Envelope proteins change each year; new strains must be selected in the first few months for manufacture



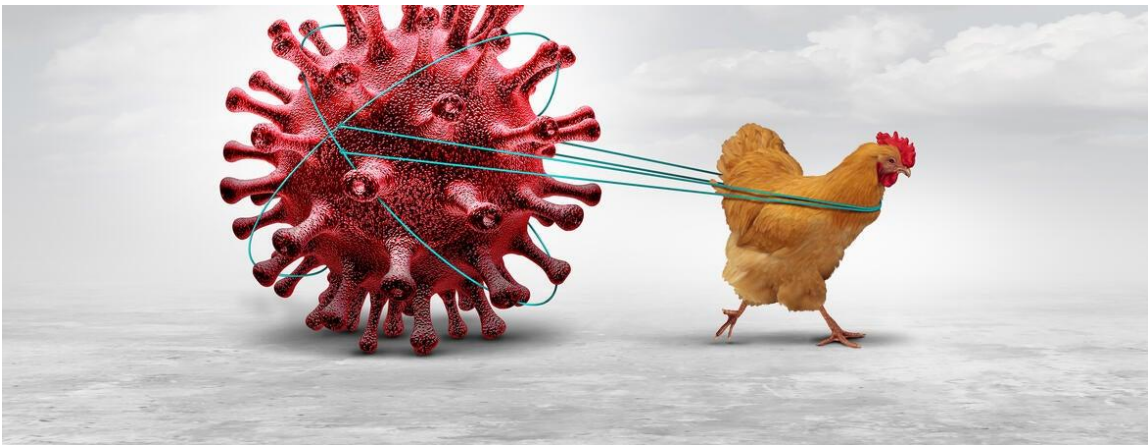
*World Health Organization Global Influenza Surveillance Network

†WHO Collaborating Centres

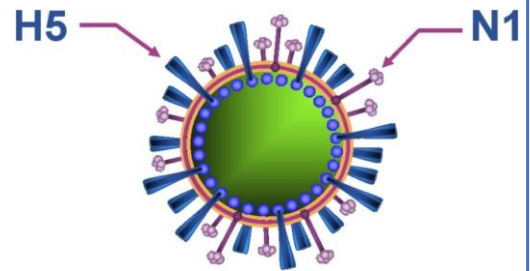
‡US Centers for Disease Control and Prevention

§US Food and Drug Administration

Avian Influenza (Bird Flu or H5N1)



- H5N1 is a subtype of the influenza A virus, commonly known as avian influenza or "bird flu".
- H5N1 is a highly pathogenic strain in birds, which means it causes high rates of death (up to 90%) when birds are infected; therefore, it is often referred to as highly pathogenic avian influenza (HPAI).
- H5N1 primarily affects birds but can occasionally infect humans and other animals.
- Spread through direct contact with infected birds, their droppings, or contaminated environments.
- H5N1 exhibits high affinity for lower respiratory tract epithelial cells in humans.
- When humans are infected, the illness can be severe, with a high mortality rate (*around 50-60% of confirmed cases*).



Recent H5N1 Cases Updates

- As of early 2025, outbreaks are ongoing in birds across Asia, Europe, Africa, and the Americas.
- Increasing reports of spillover into mammals such as sea lions, foxes, and farmed minks and cows.
- Human infections remain rare and mostly linked to close contact with infected birds.
- In North America, the majority of H5N1 cases have been in farm workers who experienced mild symptoms, and the sources of infections were cows infected with the avian influenza H5N1 virus.
- As of February 5, 2025, 67 human cases of HPAI H5N1 have been reported in 10 US states, plus one case in Canada. In most instances, symptoms have been mild with conjunctivitis being the most common symptom. Other flu-like symptoms, including cough, have been reported.
- The first death in the US of a person infected with this strain of H5N1 was reported on January 6, 2025.





H5N1 Public Concerns

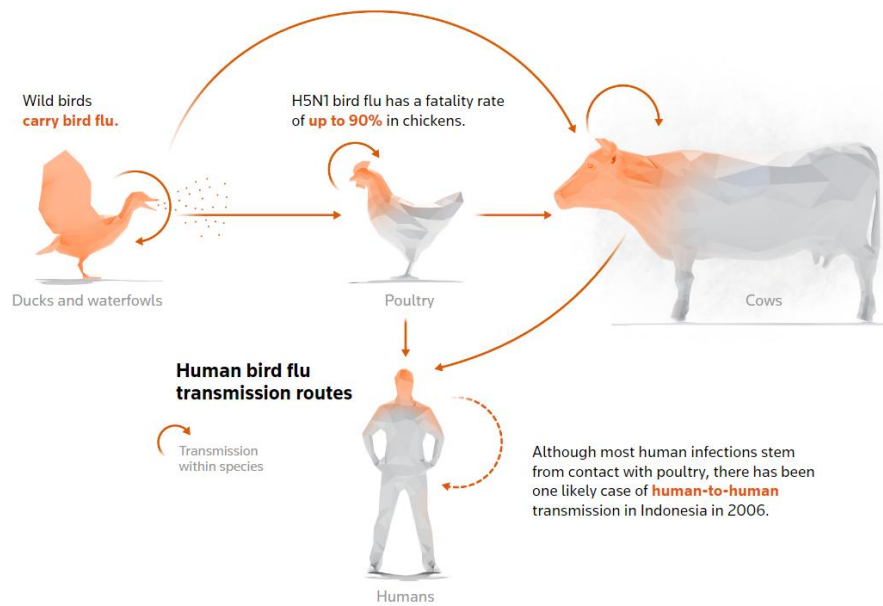
The World Health Organization and the US CDC have stated that *at this time, the avian flu risk to the general public remains low. However, epidemiologists and public health professionals are keeping a very close watch on the situation for several reasons:*

- 1- This is the first time we've found avian influenza in so many different types of mammals. This provides many opportunities for the influenza virus to mutate and/or reassort and possibly infect humans more easily.
- 2- With each human case, it is more likely that the H5N1 influenza virus will change and it's possible that these changes may lead to greater communicability or more severe disease in humans.
- 3- We are unsure how cows become infected. Each cow was either infected directly by a bird or another cow. Cows infecting other cows is concerning because the virus has adapted and can transmit from mammal to mammal. Cattle and humans frequently interact, increasing the probability of human infections.

H5N1 Public Concern

Although, ongoing genetic analysis of U.S. strains shows limited human-to-human transmission capacity, studies indicate that the H5N1 strain circulating among cows is just **one mutation** away from efficiently binding to human cells, raising concerns about potential human-to-human transmission.





Source: U.S. Centers for Disease Control and Prevention (CDC)

How common are host range jumps?



- Dead end: Very common.
- Those that produce sustaining transmission: Rare.
- Can we predict them? **No**.
- But we can know what is out there, and react (preparedness).
- We can make pan-CoV antivirals!



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HOW AI COULD HELP US PREPARE FOR THE NEXT PANDEMIC

Researchers are using machine-learning programs to predict the evolution of viruses and design vaccines.

By Ewen Callaway

In the third year of the COVID-19 pandemic, researchers reported another worrying virus. Identified in 35 people in eastern China since 2018, Langya henipavirus causes breathing problems, fever and other troubling symptoms.

Langya hasn't been linked to any deaths yet. But it is related to some deadly viruses, so researchers were keen to develop vaccines. There was just one problem: a viral protein that could form the basis of a jab seemed impossible to make in the laboratory.

"If we can't even study the protein, how are we going to understand how it works, and how

are we going to make a vaccine?" asks David Veessler, a structural virologist at the University of Washington (UW) in Seattle.

Now, thanks to artificial intelligence (AI), biologists have cracked Langya's nut. In work posted on the bioRxiv preprint server in August, Veessler's team describes using the prediction tool AlphaFold to map the structure of a protein with which Langya invades cells¹. Another AI tool identified mutations that transformed the unruly molecule into a suitable vaccine candidate.

The research, which has not yet been peer reviewed, is part of nascent efforts to use groundbreaking advances in AI, such as AlphaFold and large language models, to

Article

Learning from prepandemic data to forecast viral escape

<https://doi.org/10.1038/s41586-023-06617-0>

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 Check for updates

Nicole N. Thadani^{1*}, Sarah Gurev^{1,2,8}, Pascal Notin^{3,8}, Noor Youssef¹, Nathan J. Rollins^{1,5}, Daniel Ritter¹, Chris Sander^{1,4}, Yarin Gal² & Debora S. Marks^{1,4,10}

Effective pandemic preparedness relies on anticipating viral mutations that are able to evade host immune responses to facilitate vaccine and therapeutic design. However, current strategies for viral evolution prediction are not available early in a pandemic—experimental approaches require host polyclonal antibodies to test against^{1–16}, and existing computational methods draw heavily from current strain prevalence to make reliable predictions of variants of concern^{17–20}. To address this, we developed EVEscape, a generalizable modular framework that combines fitness



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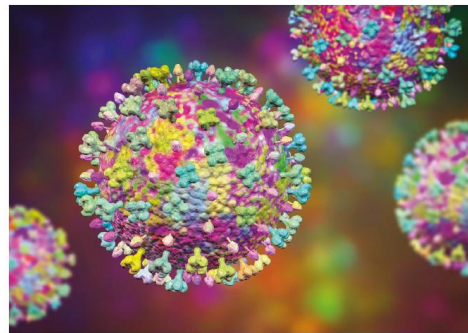
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Molecular blueprints of the Langya virus attachment and fusion glycoproteins

Zhaoguan Wang, Matthew McCallum, Liangying Yan, William Sharkey, Young-Jun Park, Ha V. Dang, Moushimi Amaya, Ashley Person, Christopher C. Broder, David Veesler
doi: <https://doi.org/10.1101/2023.08.20.554025>

Now published in *Proceedings of the National Academy of Sciences* doi: [10.1073/pnas.2314990121](https://doi.org/10.1073/pnas.2314990121)



The Langya virus (computer-generated image) is related to some deadly pathogens.

Researchers University of Washington used **AlphaFold** to predict what the virus G protein looks like because this protein can not be studied in the lab. This helps to design the vaccine.